Survey and Hardware Design of a Smart Wearable System for Pulse Wave Arrhythmia Analysis

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***Abstract*—This paper presents a survey and complete end-to-end hardware design of a compact (34 mm × 26 mm) wearable Smart Band optimized for continuous, non-invasive Pulse-Wave Arrhythmia Analysis (PWAA). The system leverages Photoplethysmography (PPG) using a high-sensitivity integrated optical sensor (MAX30102) combined with a low-power 6-axis Inertial Measurement Unit (BMI270) for adaptive motion artifact cancellation. Processing and wireless transmission are managed by the Nordic nRF52840 Bluetooth Low Energy (BLE) System-on-Chip (SoC) supported by 32 Mb external NOR Flash (W25Q32JV) for local data logging. The document outlines the continuous arrhythmia monitoring pipeline, classifies commercial devices across different product categories, and details the complete KiCad schematic, component selection, printed circuit board (PCB) floorplanning, routing, and Bill of Materials.**

GitHub Repository: Smart-Band-Design

***Index Terms*—Photoplethysmography (PPG), Pulse-Wave Arrhythmia Analysis (PWAA), KiCad PCB Design**

I. Introduction

Cardiovascular diseases are among the leading causes of death worldwide. Many heart-related disorders, such as arrhythmias, may occur without noticeable symptoms during their early stages. Continuous monitoring of heart activity can help in the early detection of such conditions and improve timely medical intervention. With recent advancements in wearable technology, smart wearable devices have become an effective solution for continuous health monitoring due to their compact size, low power consumption, and ease of use.

One of the most widely used techniques in wearable health monitoring is Photoplethysmography (PPG). PPG is a non-invasive optical method that measures changes in blood volume by illuminating the skin with light and detecting the reflected light using a photodetector. The acquired pulse wave contains valuable information about heart rate, heart rate variability, blood oxygen saturation, and pulse rhythm. By analyzing the pulse wave, irregular heart rhythms (arrhythmias) can be identified, making PPG-based devices suitable for continuous cardiac monitoring.

Several commercial wearable devices, such as WHOOP, Fitbit, Apple Watch and Samsung Galaxy Watch, use PPG sensors along with motion sensors to provide continuous health tracking. These devices integrate low-power microcontrollers, optical sensors, inertial measurement units (IMUs), wireless communication, and rechargeable batteries into compact wearable designs. Studying these devices helps in understanding current design practices and selecting suitable components for developing a custom wearable system.

In this project, a survey of commercially available wearable devices and relevant literature was first conducted to understand the principles of pulse wave arrhythmia analysis and the hardware architecture used in modern wearables. Based on this study, a compact smart wearable band was designed for pulse wave arrhythmia analysis.

II. Pulse-Wave Arrhythmia Analysis Process

A. *Optical Sensing Principles*

PPG is a non-invasive optical technique used to measure changes in blood volume in the arteries. It works by illuminating the skin with light from an LED and measuring the reflected light using a photo detector. During systole, the blood volume increases, resulting in higher light absorption and lower reflected light. During diastole, the blood volume decreases, leading to lower absorption and higher reflected light. These periodic changes form the PPG signal, which is used to measure heart rate and detect pulse wave irregularities.

Green light (520–530 nm) is primarily used for heart rate and pulse wave arrhythmia analysis because it provides a strong signal from blood near the skin surface. Red (660 nm) and infrared (880–940 nm) light penetrate deeper into the tissue and are mainly used for blood oxygen saturation (SpO_2) measurement.

The PPG signal consists of two components: the DC component (95–99)%, which represents the constant absorption by tissues, bones, and non-pulsatile blood, and the AC component (1–5)%, which corresponds to the pulsatile arterial blood flow generated by each heartbeat. Since the AC component is very small, extracting it accurately is a key challenge in PPG signal processing.

To reduce power consumption, the LEDs are driven using short pulses instead of continuous illumination. During the LED-off period, the sensor measures ambient light, which is then subtracted from the measured signal to improve accuracy and reduce the effects of external lighting.

B. Analog Front End Architecture

The electrical signal produced by the photodiode is extremely weak, typically in the picoampere to nanoampere range, and cannot be processed directly by a microcontroller. Therefore, an Analog Front End is used to condition and amplify the signal. First, a Transimpedance Amplifier converts the small photodiode current into a measurable voltage. Next, the DC cancellation circuit removes the large DC component caused by tissues and non-pulsatile blood, preventing the amplifier from saturating and allowing the small pulsatile signal to be extracted. The remaining AC component is then amplified by a Programmable Gain Amplifier to make full use of the input range of the ADC. Finally, the ADC converts the conditioned analog signal into a high-resolution digital signal, typically with 18–20 bit resolution and sampling rates between 50 Hz and 400 Hz. The digitized PPG signal is then processed by the microcontroller for heart rate measurement and pulse wave arrhythmia analysis.

C. Digital Filtering and Motion Artifact Rejection

The raw PPG signal obtained from the optical sensor is often affected by various sources of noise, including electronic interference, ambient light, and motion artifacts caused by wrist movements. These unwanted signals can distort the PPG waveform and reduce the accuracy of heart rate and arrhythmia detection. To improve signal quality, the acquired PPG signal is first passed through a bandpass filter with a frequency range of approximately 0.5 Hz to 5 Hz, which corresponds to the normal range of human heart rates. This filter suppresses low-frequency baseline drift and high-frequency noise while preserving the useful cardiac information.

Motion artifacts are one of the most significant challenges in wearable PPG systems. During hand or wrist movements, the deformation of skin and underlying tissues changes the optical path between the LED and the photodiode, producing fluctuations that can closely resemble genuine heartbeat signals. To distinguish these false signals from actual pulse waves, an accelerometer continuously measures the movement of the wrist. The motion data is then used by signal processing algorithms to identify and estimate the motion-induced noise. An Adaptive Least Mean Squares (LMS) filter models the motion artifacts using the accelerometer data as a reference and continuously subtracts the estimated noise from the PPG signal. This adaptive filtering process significantly improves the quality of the PPG signal, enabling more accurate heart rate measurement and pulse wave arrhythmia analysis even during moderate body movements.

TABLE I
ECG VS PPG

ECG	PPG
Measures electrical activity	Measures blood volume changes
Requires Electrodes	Require LEDs and Photodiodes
Higher Diagnostic accuracyr	Good for continuous screening
More expensive	Lower Cost
Clinical Equipments	Wearables

D. Feature Extraction

The next step is to identify the systolic peak in each cardiac cycle of the PPG signal. The systolic peak corresponds to the maximum blood volume in the arteries during heart contraction and is used to calculate important cardiovascular parameters. Peak detection is commonly performed using a threshold-and-slope algorithm, in which the signal must exceed an adaptive threshold and simultaneously be a local maximum. To prevent the same heartbeat from being detected multiple times, a refractory period of at least 300 ms is maintained between successive peaks. Once the systolic peaks are detected, the Inter-Beat Interval (IBI) is calculated as the time difference between two consecutive peaks. The IBI is then used to determine heart rate, heart rate variability, and detect irregular pulse rhythms associated with arrhythmias. The processed data is managed by the SoC, which controls sensor operation, performs signal processing, manages power consumption, and transmits the measured data wirelessly to a smartphone or other external device using BLE. This integrated approach enables continuous, real-time monitoring while maintaining low power consumption, making it suitable for wearable health monitoring applications.

III. SURVEY OF COMMERCIAL DEVICES UTILIZING PWAA

A. Smartwatches

Smartwatches are the most popular category of wearable health monitoring devices. They combine powerful processors, wireless connectivity, optical sensors, and advanced software to provide continuous monitoring of cardiovascular health, physical activity, and sleep.

- **Apple Watch Series 9 / Ultra 2:** Uses multiple green, red, and infrared LEDs with photodiodes for continuous PPG monitoring. It detects irregular heart rhythms and supports on-demand single-lead ECG recording for atrial fibrillation (AFib) detection.
- **Samsung Galaxy Watch 7 / Ultra:** Features Samsung's BioActive sensor, which integrates PPG, ECG, and Bioelectrical Impedance Analysis (BIA). It provides continuous heart rate monitoring, AFib detection, blood oxygen measurement, sleep tracking, and body composition analysis.
- **Google Pixel Watch 3 / 4:** Combines Fitbit's health monitoring algorithms with multi-path PPG sensors to provide continuous heart rate monitoring, sleep analysis, and AFib detection.
- **Garmin Venu 4 / Fenix 8:** Equipped with Garmin Elevate optical heart rate sensors for continuous heart rate monitoring, Pulse Ox (SpO_2), stress monitoring, sleep analysis, and heart rate variability (HRV). These watches are optimized for fitness, outdoor activities, and long battery life while supporting advanced physiological monitoring.
- **Amazfit Balance and Huawei Watch GT4:** Utilize multi-channel optical PPG sensors for continuous heart

rate, blood oxygen, stress, and sleep monitoring while maintaining extended battery life.

B. Smart Bands

Screenless smart bands are designed for continuous physiological monitoring without a display. Eliminating the display reduces power consumption, decreases weight, and enables multi-day battery life, making them suitable for 24/7 health monitoring.

- **WHOOP MG (Medical Grade):** A medical-grade wearable focused on continuous cardiovascular monitoring, recovery analysis, and strain measurement. It employs multiple green, red, and infrared LEDs with multiple photodiodes to acquire high-quality PPG signals and stores data locally before synchronizing with a smartphone via Bluetooth Low Energy (BLE).
- **WHOOP 4.0 / 5.0:** Designed for continuous heart rate, recovery, sleep, and activity monitoring using multi-wavelength PPG sensors and onboard memory for uninterrupted data logging.
- **Hume Band 2.0:** A lightweight screenless wearable that continuously monitors heart rate, heart rate variability, stress, recovery, sleep quality, and daily activity while providing long battery life.
- **Huawei Band 9 / Honor Band 10:** Affordable fitness bands equipped with optical PPG sensors for continuous heart rate, SpO_2, sleep, and stress monitoring. They offer slim designs, long battery life, and comprehensive health tracking features.

C. Hybrid Smartwatches

Hybrid smartwatches combine the appearance of traditional analog watches with modern health monitoring capabilities. Hidden displays and optical sensors provide continuous health tracking while maintaining long battery life.

- **Withings ScanWatch 2, Nova, and Nova Brilliant:** Provide continuous heart rate monitoring using multi-wavelength PPG sensors, on-demand ECG recording, blood oxygen measurement, and up to 30 days of battery life.

D. Smart Rings

Smart rings provide continuous physiological monitoring in a compact and lightweight form factor. Since fingers contain dense capillary networks, they offer excellent PPG signal quality for cardiovascular monitoring.

- **Oura Ring Gen 4 and Samsung Galaxy Ring:** Integrate green, red, and infrared LEDs, temperature sensors, and motion sensors for monitoring heart rate, sleep, blood oxygen saturation, body temperature, and activity.
- **CART-I Ring (Sky Labs):** A medical-grade smart ring designed specifically for continuous atrial fibrillation (AFib) monitoring and remote patient care.

E. Medical Patches

Medical patches prioritize diagnostic accuracy over portability and are widely used for long-term clinical ECG monitoring.

- **iRhythm Zio XT:** A waterproof adhesive chest patch capable of recording continuous single-lead ECG signals for up to 14 days. The recorded data is analyzed using AI-based algorithms for arrhythmia detection.
- **AliveCor KardiaMobile 6L and Spyder Patch:** Portable ECG devices that provide real-time wireless arrhythmia monitoring through smartphone applications.

F. Smart Textiles

Smart textiles integrate conductive electrodes into garments, enabling comfortable and continuous physiological monitoring during daily activities and sports.

- **Hexoskin Medical System, BioSerenity Cardioskin, and Nuubo nECG Holter:** These smart garments provide continuous ECG monitoring, respiration analysis, activity tracking, and long-duration physiological monitoring for clinical and sports applications.

IV. SMART BAND KICAD DESIGN

To demonstrate the practical realization of a compact PWAA monitor, a custom PCB was designed in KiCad. The board measures 34 mm × 26 mm with side strap cutouts, adopting the screenless Whoop 4.0 form factor.

A. Root Schematic Architecture

The proposed smart wearable band is designed using a modular hierarchical architecture in KiCad, where each functional block is implemented as a separate schematic sheet and integrated into the root schematic. The system is powered through a USB Type-C interface, which provides a 5 V input supply for battery charging and supports USB communication with the microcontroller. The USB Type-C connector uses $5.1\,\mathrm{k}\Omega$ pull-down resistors on the CC1 and CC2 pins for device detection, while the D+ and D- lines are connected directly to the microcontroller for programming and debugging. A Microchip MCP73831 linear battery charger is employed to charge a single-cell Li-ion battery using the constant-current/constant-voltage charging method, with the charging current set to approximately 500 mA using a $2\,\mathrm{k}\Omega$ programming resistor. The rechargeable 150 mAh Li-Po battery can be connected through a compact JST GHS two-pin connector, making the design suitable for wearable applications.

The battery voltage, which varies between 3V and 4.2V, is regulated to a stable 3.3 V using the TPS7A0233 low-dropout voltage regulator. This regulated supply powers all digital and sensing components while maintaining extremely low quiescent current, thereby extending battery life. The central processing unit of the system is the Nordic nRF52840-QIAA-R System-on-Chip, which integrates a 32-bit ARM Cortex-M4 processor operating at 64 MHz along with Bluetooth Low Energy (BLE) connectivity. The microcontroller is responsible for sensor interfacing, signal processing, power management, local data handling, and wireless communication with external devices. To enable offline storage of physiological data, a W25Q32JV 32 Mbit Serial NOR Flash memory is connected through the SPI interface, allowing sensor data to be stored when a Bluetooth connection is unavailable.

Physiological signals are acquired using the MAX30102 optical PPG sensor, which integrates red and infrared LEDs, a photodetector, an analog front end, and an 18-bit analog-to-digital converter within a single package. The sensor communicates with the microcontroller through the I²C interface and uses an interrupt pin to indicate the availability of new measurement data. To improve measurement accuracy during body movement, the design also includes the Bosch BMI270 six-axis inertial measurement unit, which provides three-axis accelerometer and three-axis gyroscope data. The motion information obtained from the IMU is used for motion artifact reduction, thereby improving the reliability of pulse wave analysis. The combination of low-power components, efficient power management, wireless connectivity, and compact PCB design makes the proposed hardware architecture well suited for continuous pulse wave arrhythmia monitoring in a wearable smartband.

B. PCB Layout and Physical Floorplanning

The component placement follows a functional floorplanning approach to minimize routing complexity and optimize system performance. The Nordic nRF52840 SoC is positioned near the center of the top layer to reduce trace lengths to peripheral devices. The USB Type-C connector is placed along the top edge of the board for convenient charging and programming access, while the W25Q32JV external flash memory and BMI270 motion sensor are positioned close to the microcontroller to simplify high-speed SPI and I²C routing. The battery charging circuit, consisting of the MCP73831 charger and TPS7A0233 LDO regulator, is located together to provide a compact power management section. The JST battery connector and reset push button are placed near the bottom edge for easy battery connection and system reset. Four plated mounting holes are provided at the corners of the PCB to securely fasten the board within the enclosure.

The bottom layer is reserved exclusively for the MAX30102 optical PPG sensor, which is positioned at the geometric center of the PCB. This placement ensures direct and uniform contact between the sensor and the user's wrist when the device is worn, thereby improving the quality of the acquired photoplethysmography signals. The PCB outline incorporates rounded corners with a 2 mm radius to improve user comfort and reduce sharp edges. In addition, two rectangular slots are provided along the left and right edges of the board, allowing a fabric wrist strap to pass directly through the PCB. This mechanical arrangement provides a secure attachment mechanism while maintaining a slim and lightweight wearable design.

V. Results

The completed PCB design was successfully verified using the KiCad Design Rule Check (DRC) tool. The verification

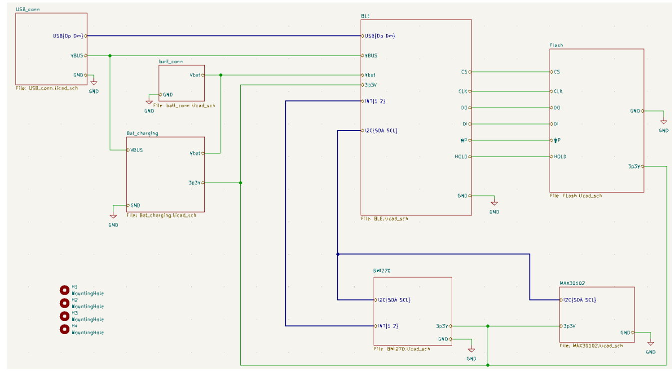

Fig. 1. Root Schematic

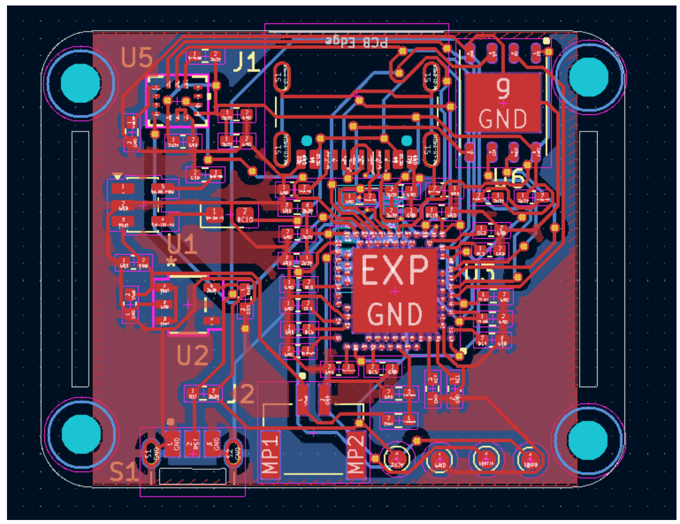

Fig. 2. PCB Layout

confirmed zero design rule violations and zero unrouted nets, indicating that all electrical connections were correctly routed and the PCB satisfies the specified design constraints.

After successful verification, all manufacturing files required for PCB fabrication and assembly were generated. The Gerber fabrication files include the copper layers, solder mask layers, silkscreen layers, and board Edge Cuts, which are used by PCB manufacturers to produce the printed circuit board. n addition, a Bill of Materials (BOM) file was generated, listing all components along with their manufacturer part numbers and quantities required for assembling the smart wearable band.

VI. Conclusion

The estimated hardware cost of the proposed smartband is approximately $Rs.$4,000, including the major electronic components, passive components, PCB fabrication, connectors, and battery. Although this cost is comparable to the manufacturing cost of several commercial wearables, it is significantly lower than the retail prices of premium products such as Whoop, Oura Ring, and high-end smartwatches. Commercial wearable devices are priced much higher because

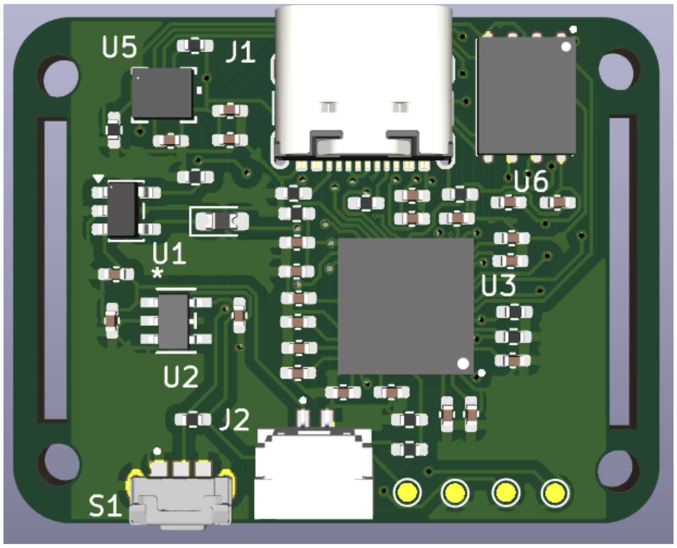

Fig. 3. 3D Model Top View

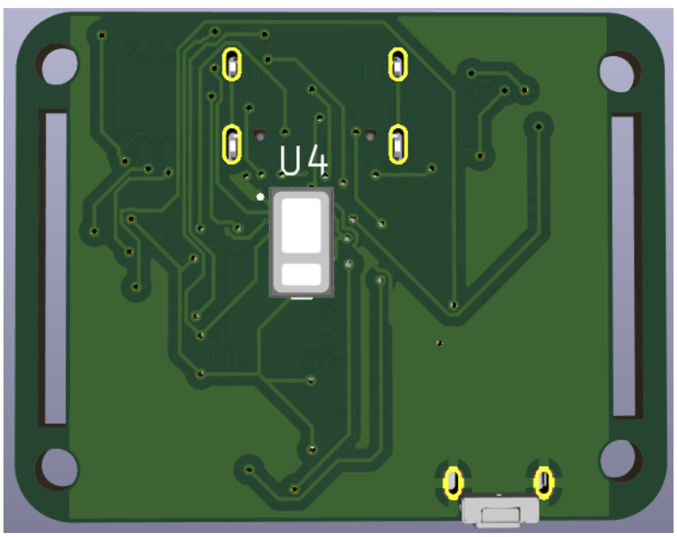

Fig. 4. 3D Model Rear View

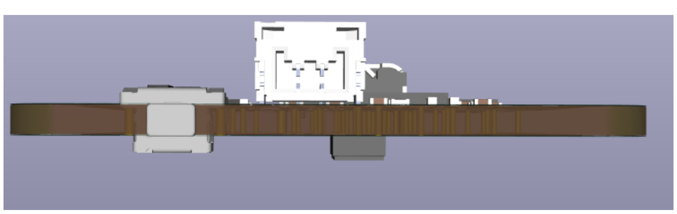

Fig. 5. Side view showing Reset Button

TABLE II
MAJOR COMPONENTS USED IN THE SMART BAND

Component	Part Number	Approx. Price (INR)
Nordic BLE SoC	nRF52840-QIAA-R7	734
Optical PPG Sensor	MAX30102EFD+T	1306
6-Axis IMU	BMI270	404
Serial NOR Flash	W25Q32JVZPIQ	224
Li-Ion Battery Charger	MCP73831-2-OT	73
3.3 V LDO Regulator	TPS7A0233DBVR	81
USB Type-C Receptacle	TYPE-C-31-M-12	250
Battery Connector	SM02B-GHS-TB	34

the final product cost includes research and development, proprietary algorithms, custom mechanical enclosure design, mobile application development, cloud infrastructure, certifications, quality assurance, marketing, distribution, customer support, and company profit margins. In contrast, the proposed design focuses solely on the essential hardware required for pulse-wave arrhythmia monitoring, resulting in a much more economical solution while maintaining the core sensing and wireless communication capabilities.

This smartband can be further enhanced by implementing embedded firmware for real-time PPG signal acquisition, digital filtering, systolic peak detection, and arrhythmia classification. A dedicated mobile application can be developed to display live physiological data, maintain health records, and generate alerts for abnormal heart rhythms. The hardware can also be expanded by integrating additional sensors such as an ECG front-end, skin temperature sensor, galvanic skin response (GSR) sensor, or blood pressure estimation algorithms to provide comprehensive health monitoring. Incorporating machine learning models for automatic arrhythmia detection and personalized health analysis can further improve diagnostic accuracy. Future versions may also include wireless charging, a flexible PCB, waterproof enclosure, improved power management for longer battery life, and cloud-based data synchronization for remote patient monitoring and telemedicine applications.

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